

Oral Exam Questions

Fall 2023

Directions: The following is a list of questions. From this list, Dr. Owad will pick one question for you to solve and give an explanation of your work. You will have 20 minutes, starting at the time you selected. See the current week of our Blackboard page for the time that you are registered for. To prepare, you should be sure you can solve each of the following questions. Make sure you can explain your reasonings. You are welcome to prepare for these with other students, your notes, the book, and me. You may bring notes with you to aid you.

1. Give a combinatorial proof for the identity $1 + 2 + 3 + \cdots + n = \binom{n+1}{2}$.
2. Consider an equilateral triangle with 7 distinct points on one side, 8 on another, and 9 on the last side. None of these points are on the corners of the equilateral triangle. How many nondegenerate triangles are there with vertices on the points.
3. If x and y are even and z is odd, show that $xy + z^2$ is odd.
4. Let $a_n = 6n + 2$, and $b_n = \sum_{i=1}^n a_i$, what is b_{100} ? Find a closed formula for b_n . Prove your formula works with induction, for $n \geq 1$.
5. Let $a_n = \frac{2}{3^n}$, and $b_n = \sum_{i=0}^n a_i$, what is b_{25} ? Find a closed formula for b_n . Prove your formula works with induction, for $n \geq 0$.
6. Consider the following sequence of hexagons for $n = 1, 2, 3$. Let h_n be the number of hexagons in each diagram. Produce a recursive formula and a closed formula for h_n .

